ON Earth Life

I

atomic structure of reality

what it realy is

water

why everything needs it
how it affects the energy levels
how is it managed by the world

plastic

ideas and issues

what can you do

the extra pages with explanations are only in the first topic

Reality is layered..
And what both Einstein and Stephen Hawking proved - there is no easy and quick way to understand it.

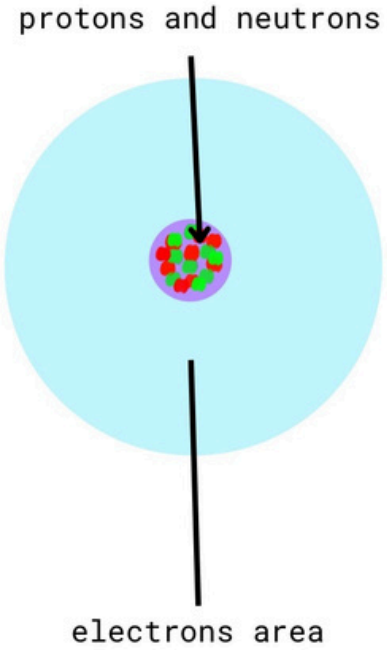
First layer, known by many, is the **atomic structure of reality**. It really is not exactly what it seems like, but through the publications we will dig deeper, starting from this level.

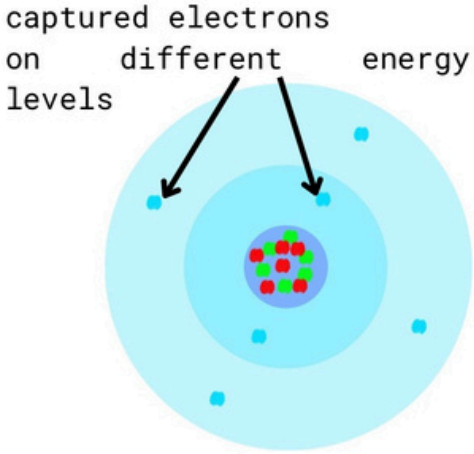
This type of knowledge seems absolutely abstract, even though it describes the reality.
But the more puzzles of it you get, the more sense it starts to make.

You can treat this level as facts.
The answers to further questions will be there in another on Earth Life magazine parts.

Most of the ordinary matter around us is made of numerous connected (or no) **atoms**. Every atom in most cases contains 3 specific particles - protons, neutrons and electrons, arranged in a specific way.

Protons and neutrons form a dense nucleus in the center, while electrons are found around it.





singular carbon atom

- 6 protons
- 6 neutrons
- 6 electrons

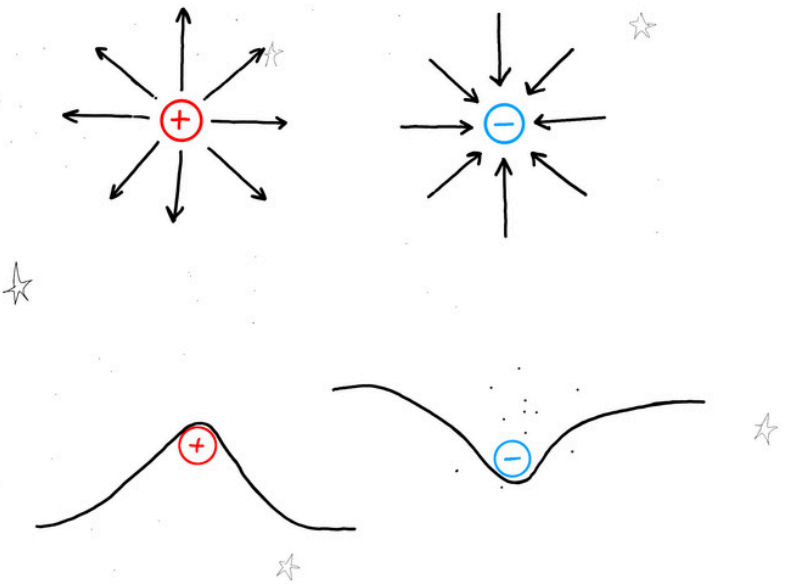
every physical thing around you is made of atoms - 3 connected particles.
Atoms connect in enormous amounts and ultimately create things around you (our planet, your body, your cat etc.)

We differentiate atoms with different amounts of protons in the nucleus into elements. All elements which we know are described in Mendeleev's periodic table.

Every particle has energy.
Not every particle has an electric charge.

Electric charge determines how a particle interacts with the electromagnetic field.

(about fields, energy and other crucial particles like light particles in another publication)



From what we know - electric charge can be positive, negative, or zero.

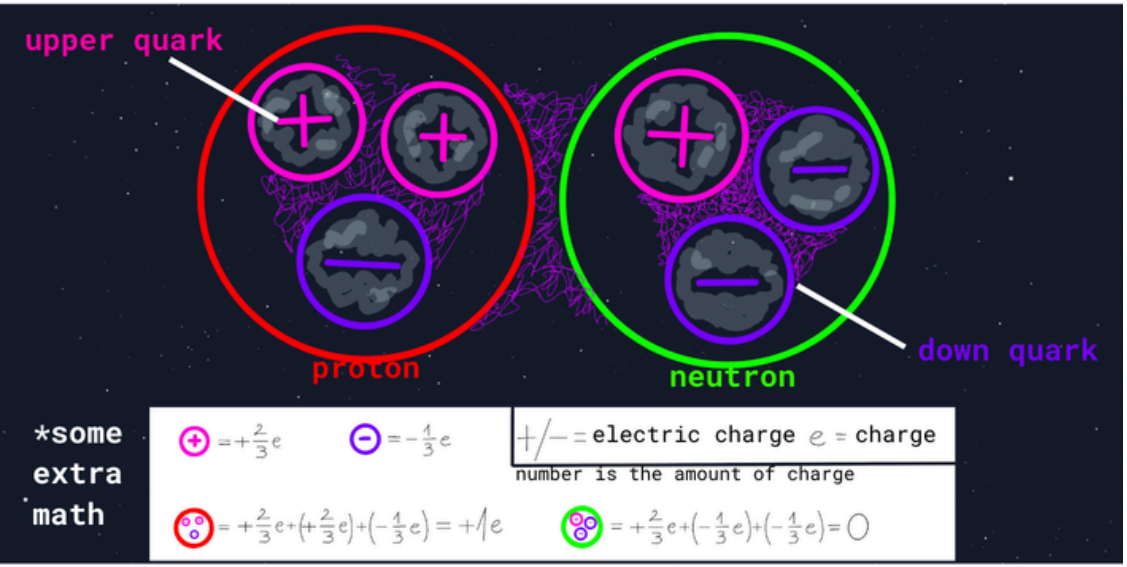
The same charges repel each other and different charges attract.
They also react differently with the field (as shown above).

However, except for these characteristics, we still have no idea why the differences between these charges exist, because they are structured the same way.

Inside a proton and a neutron, there are other particles - quarks that are held together by the strong interaction. This interaction is mediated by other particles called gluons.

Quarks have electric charges.

A proton has a net positive electric charge, while a neutron has no net electric charge.



The strong interaction (provided by gluons) is much more important than the electromagnetic repulsion of the similarly charged quarks at the extremely small distances inside a proton or neutron. So it keeps the quarks bound together. Gluons also provide the force to keep protons and neutrons together in the form of a nucleus.

Every proton and neutron have one fixed amount of energy.

two atom particles - protons and neutrons - contain another three particles in them - quarks

Both these prticles inside protons and neutrons and the nucleus (made of protons and neutrons) are held together by strong forces mediated by another particles - gluons.

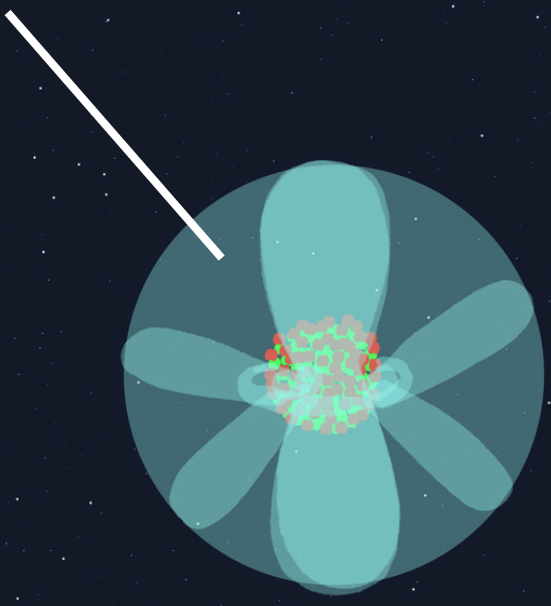
this force is needed, because for exaple protons have the same positive electric charge, therefore naturally they repell one another.

In an atom, electrons carry negative electric charge and, as far as we know, they are not made of any other, smaller particles*. But electrons don't have just one fixed amount of energy. They can exist in certain, different energy states. You can think of these states as different energy levels. And each level can hold only a limited number of electrons.
*electrons and quarks (and some others) are fundamental particles.

But an electron in one of these energy levels does not sit in one particular spot, like a tiny ball sitting at a specific point. Quantum physics tells us that, before we measure its position, we cannot say exactly where the electron is. Instead, we can describe

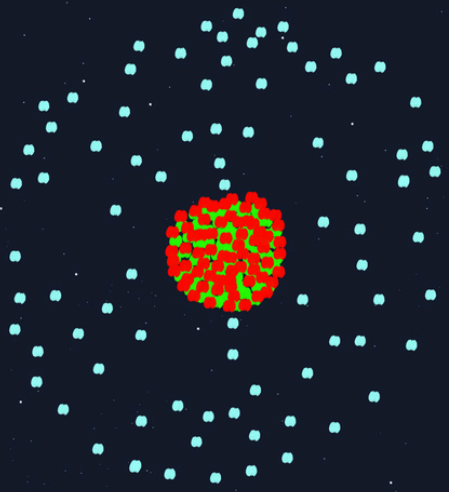
the probability of finding it in different places around the nucleus.

the area isn't necessarily round



example of captured gold atom

- **79 protons**
- **79 neutrons**
- **79 electrons**



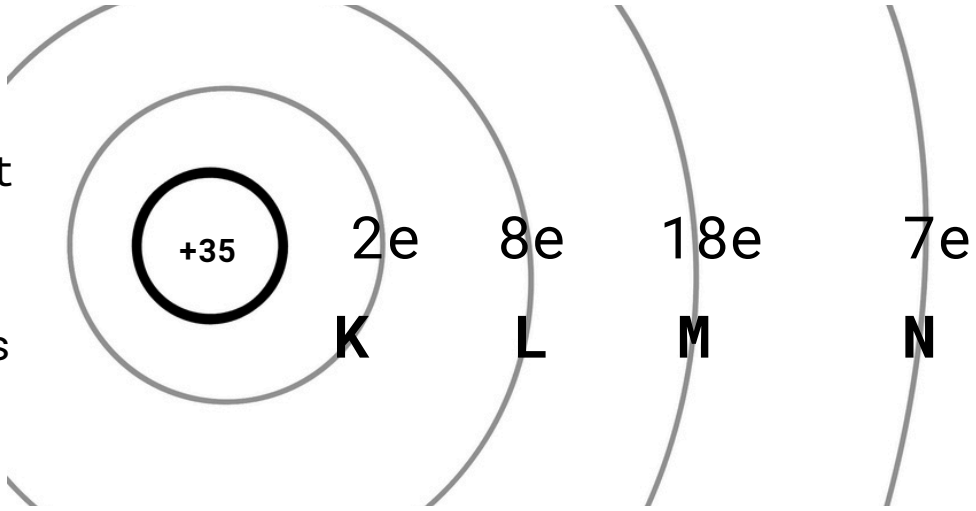
electrons on each energy level look a little different and has a higher chance to be found in different areas

electrons are carrying negative electric charge.

They are somewhere aroud the nucleus, but they don't ocupy specific place.
They occupy specific and limited energy level.
And specific energy level provide different areas around nucleus where electrons are most likely to be found.

This popular explanation from school is widely misunderstood:

bromine element
- 35 protons
- 35 neutrons
- 35 electrons



These are not layers explaining where and how many electrons are there. This only explains the energy levels and how many electrons can be on each energy level.

Before measuring its position, we cannot assign the electron to a single definite location; instead, we can only describe the probability of detecting it at different locations and the location does not need to be directly on orbits like above.

And science can’t describe yet how exactly this uncertainty influences reality.

We can tell how many electrons are in an atom and therefore how many electrons are in each energy level (but still… we cannot define their exact location until we somewhat capture them).

the usual school explanation of atom is explaining how many electrons can be there on each energy level rather than the literall structure of an atom and places where the electrons are

Atoms connect together by ‘connecting’ the outer parts - electrons.

After connecting, particular electrons occupy quantum states (energy levels) that extend over two or more atoms.

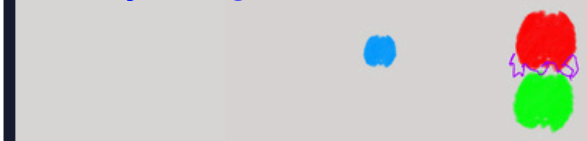
After forming H2O bonds, each hydrogen can now have access to two electrons in its first energy level instead of one, while the oxygen can now have access to eight electrons in its second energy level instead of six.

The electrons are shared between the atoms rather than belonging exclusively to one atom. In this way, each hydrogen reaches a full first energy level, while oxygen reaches a full second energy level.

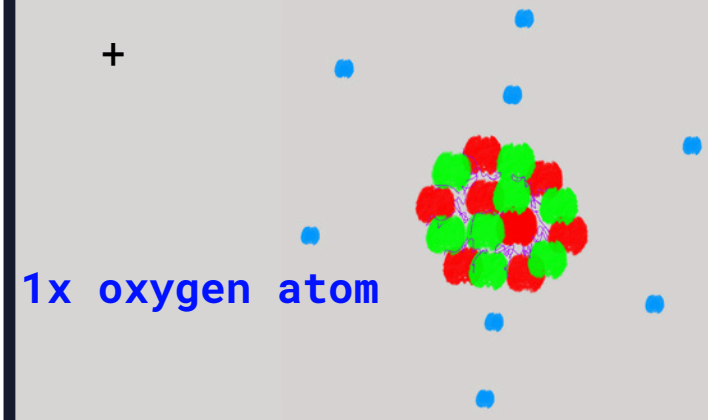
This is because, in each atom, the maximum number of electrons in the first energy level is 2, and in the second energy level it is 8.

example : water molecule

2x hydrogen atom



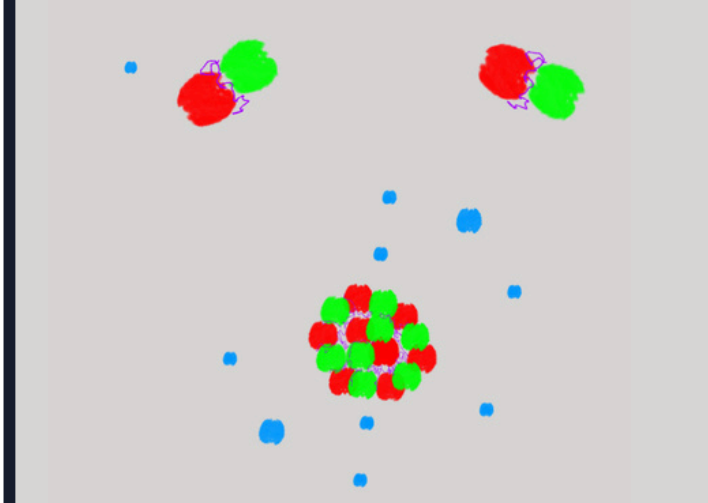
+



1x oxygen atom

=

1 water molecule



atoms connect together by opening their electron fields for electrons from another atom with which they connect. So after they connect they share electrons rather than sticking any parts.

&

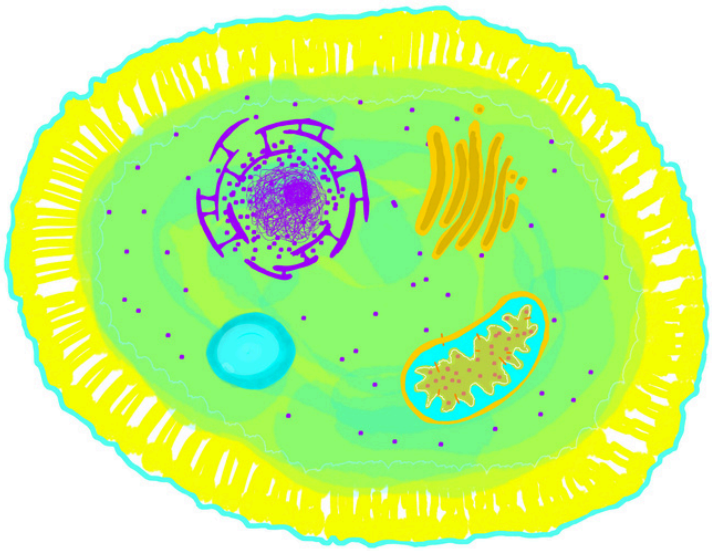
SPEAKING OF WATER AND ENERGY



We originally come from water.

We - meaning every living creature.

Every living creature - meaning us, plants, mushrooms, other animals - everything made of cells.



cells are of course made of atoms arranged in a very specific structures, which can somehow obtain and transform energy from the environment. Every organism is made of numerous cells. Some organisms like bacterias are made of singular cells.

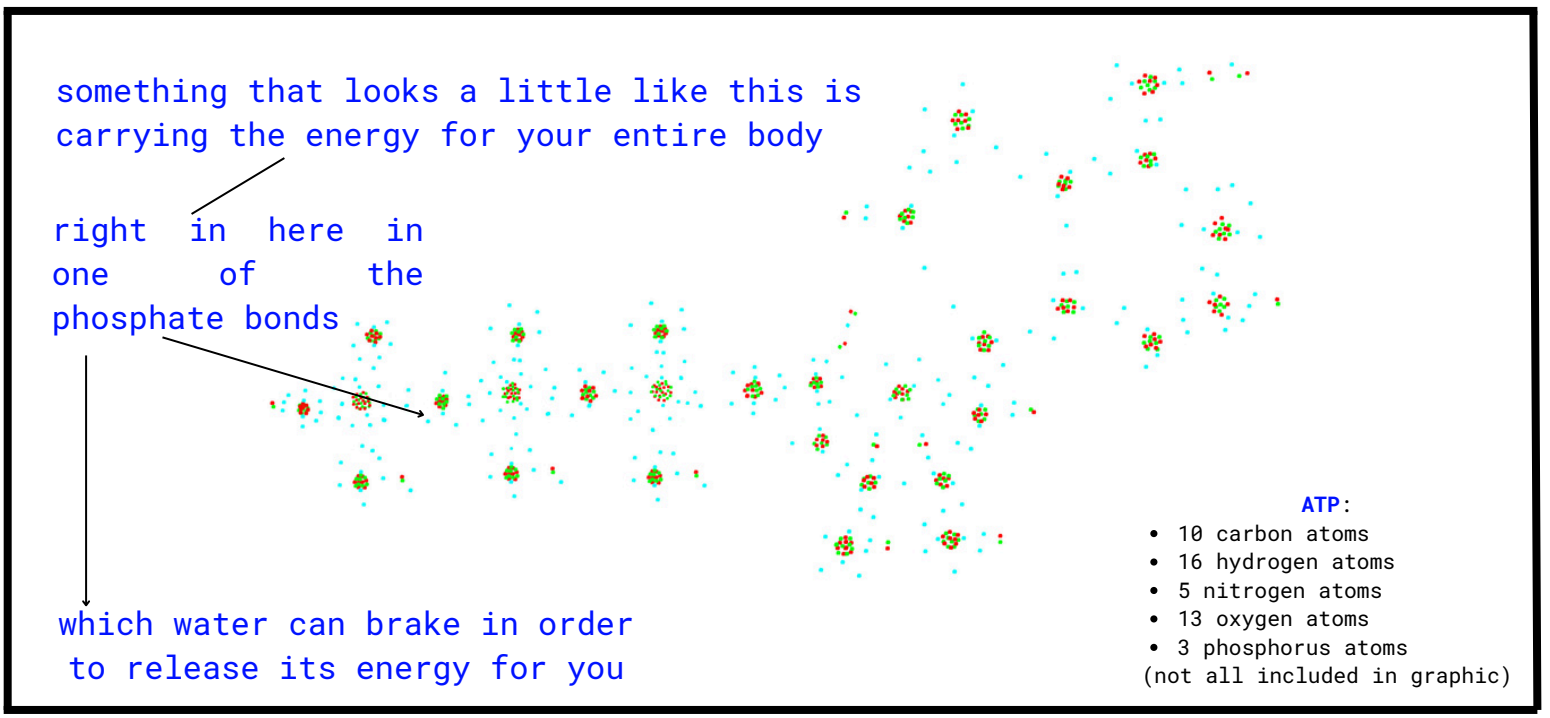
There is no living creature on Earth that doesn't need water in order to live. Life first arranged itself in water.

Water provides the environment for our cells and participates in many of the biochemical reactions that for instance allows cells to extract the energy (for us) from food.

Water is the environment of every cell in our body. Basically, we are made of cells that evolved in and still to this day, need to live in a water-rich environment. BUT together they build creatures that can live on land - us (and every other on land life - plants, animals, etc.)

Water participates in the reactions that allow ATP* to be used to power almost every cellular process (walking, breathing, thinking.. basically everything).

*ATP - our body's universal energy carrier molecule

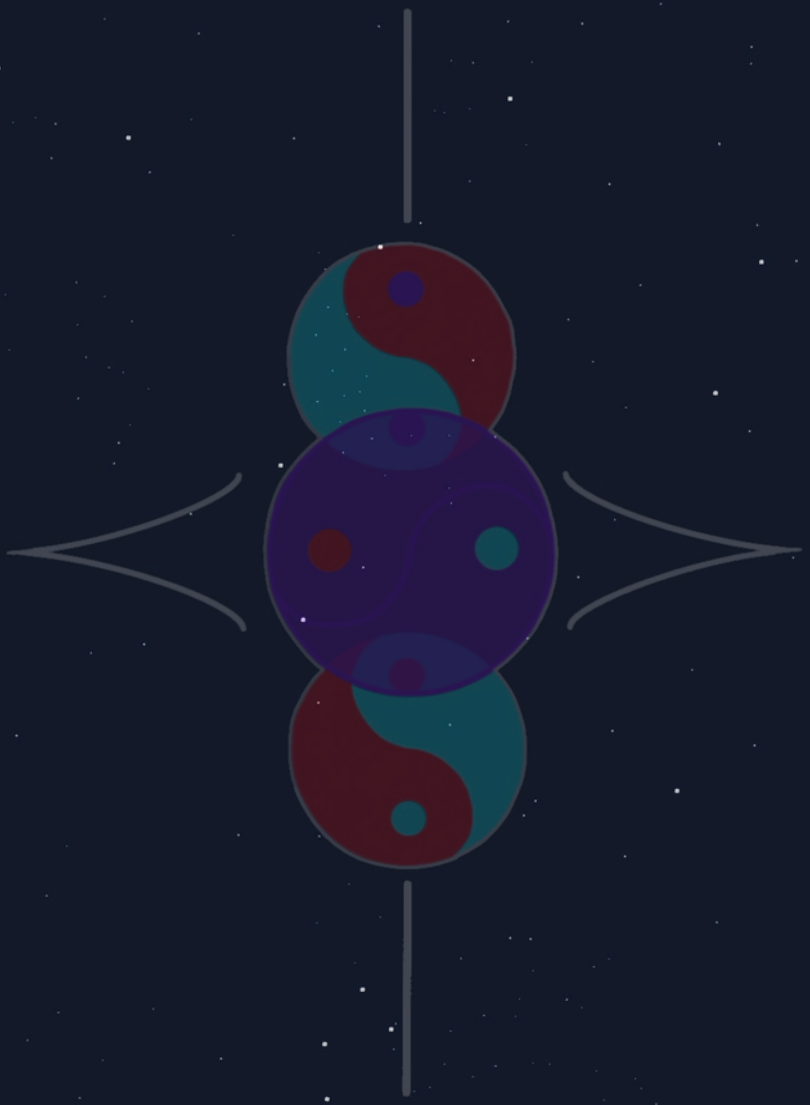


During photosynthesis, plants split water into oxygen, protons (hydrogens without electron) and electrons. The oxygen is released into the atmosphere, while the electrons allows plants to continue the photosythesis reaction.

Water also affects how much energy our body has to spend to maintain normal physiological function - for instance, when you are dehydrated, your blood plasma volume decreases and your heart may have to work harder to maintain adequate blood flow. It needs more energy to pump thicker blood.

Taking care of your body means taking care of your cells. You can stop caring about them once you learn how to consciously control every organs' work in your body at the same time (as your cells do it for you). Until then, it's better to take care of them, so they can do the complicated work for you. You can support them by giving them enough water and not giving them much extra work like detoxifying whatever toxic substances you put in your body. Because it's their extra work. And both

their energy production and usage equals your energy level.



How big companies are obtaining water:

Most countries obtain water from underground sources, which can be very harmful for local vegetation. Mainly because it decreases water levels below the ground. And plants mainly absorb water through their roots. So now, in order to obtain water and therefore to live, plants need to grow deeper roots. However, plants usually can't keep up with changes in water levels. And so high chances are they die.

And the death of a thousand trees doesn't even make economic sense.

First of all, it destroys the natural cycle which could help us to change the way we obtain water to a less depleting and cheaper one. That is because trees and other plants release water into the atmosphere through their leaves. And by obtaining water through their roots, they basically act like a natural pump.

the accumulation of evaporated water in the right circumstance takes a big part in forming rain clouds.

Then, of course, rain comes, rehydrates the soil and the cycle can start again. So it can be very much endless.

Therefore, the death of vegetation around water mines makes it harder for water levels to recover, and we need to dig deeper and deeper and eventually switch locations.

LUCKILY

There are some countries which noticed that working with natural forces is easier and, overall, cheaper than fighting against them. But most countries seem like they cannot change to a better way only because they have been doing otherwise for some time already. Which is clearly illogical.

Some countries have strict rules for monitoring rivers or act strongly towards depolluting them, which makes it possible to collect water from lakes and rivers. Which with right management can be less depleting than collecting underground water, because it doesn't decrease the water levels to the point at which plants, in order to survive, need to grow bigger roots.
This also allows natural cycles to be continued and supports the overall cleanliness, which in turn increases the amount of other usable water resources (for example, fish) and allows tourism to develop, which all together can add extra income for the country.

- **Norway**
- **England**
- **Canada**

Some countries have developed infrastructure which allows them to collect rainwater from both urban and natural areas, thanks to retention tanks, which collect the rainwater and then convey it via underground infrastructure to a treatment facility. And after it passes certain tests, it is made available for people to use, for example as drinkable tap water.

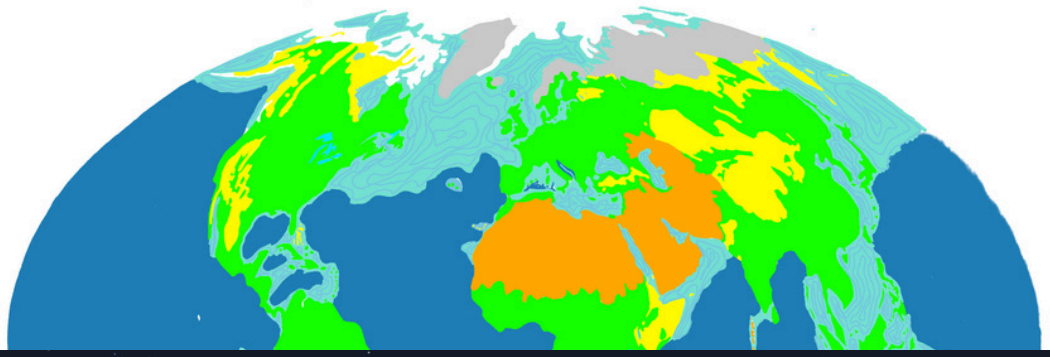
- **Singapore**
- **Holland**
- **New Zealand**

And some countries have developed desalination of seawater to the point where it can be drinkable.

- **Saudi Arabia**
- **Emirates**

About extra incomes:
all countries which have independent and non-destructive ways of collecting water are less dependent on more expensive water imports from other countries.

Also, since we have a free market, there are no stops for people who can, to invest in poorer countries with rainy tropical climates, to build optimal rainwater collection points, which surely can satisfy both the investors and the countries receiving the investment.



There is an absolute need for a global water treatment agreement. Not only because it's crucial for us to live, but also because one country's actions regarding its' waters are directly influencing every other country's waters. Mainly because of forces we can't control, like for example water tides caused by the rotation of the Earth.
So if one country pollutes its' waters, it will for sure affect other waters (once polluted rivers reach the oceans).

ALSO one more issue (since everything is made of atoms that are not capable of creating forever-lasting bonds) is how we dispose the water. This issue was once well managed, so there is no need to investigate the best solution.

PLASTIC..

Plastic on its molecular level is porous and absolutely nothing can change that.

So small plastic particles - micro- and nanoplastics - can easily detach from the main plastic element through its' abrasion and aging. They occur in our water (because of plastic bottles and pipes) and they occur in our air, therefore in our lungs (because of plastic dust, made from plastic clothing).

So nowadays we inhale and consume plastic quite directly.
Of course, it was not always like that (widely used plastic has been with us for around 70 years, and therefore our cells definitely didn't evolve to easily detoxify it or digest it (and probably never will)).

Brief and simplified history of how water was distributed:

3000 B.C.E - 1920 A.D.
Water was transported in lead pipes into ceramic, glass or other vessels.
It lasted until science was developed enough to state without a doubt that water in which lead is present has harmful effects on health.

1920 (and earlier) A.D. - 1980 A.D.
Water was transported in stainless steel pipes into glass bottles.
It lasted until a cheaper and lighter material - plastic - was introduced.

1980 A.D. - now
Water is transported in plastic pipes into plastic bottles.

And now science is developed enough to state almost certainly that plastic occurring in our bodies is harmful.
(more details in another free and easy article in shorter form, available on www.oelfvmag.com)

So there really only is a need for one more change.

Of course, the issue is now on a bigger scale than ever before, but we surely are more developed than we were ever before.



***The idea*:**

1. Destroy underground water wells and collect their materials to build new stainless steel water distribution pipes and new water collection systems.
2. Build new water collection systems and change plastic pipes to stainless steel ones.
3. Before that, if needed, invest in cleaning groundwater (which will pay for itself through both increased tourism and water independence).
4. Use plastic collected from old plastic bottles and pipes (and maybe also from water and land depollution) as building materials (ideas later).
5. Distribute water in stainless steel containers and later devide it into glass bottles, to avoid losses during transport.

From a single-person perspective, it seems impossible. But from the perspective of united big companies and governments, it seems like a doable, long-term and logical plan.

And since we know that plastic is harmful, not changing plastic infrastructures is an **obvious action against humanity.**

As a single person, what you can do is obtain a cute, multi-use water bottle. It brings more change than you can imagine. And it can be a useful personal belonging.

Collected plastic from old bottles and pipes can be reused as a material for many different things. For example, **cool bins**. In a quantity that will make littering illogical and with a design that:

- makes everyone know what kind of trash they are for
- prevents bad smells
- doesn't destroy the architectural concept
- is easy to pick up

or we can reuse plastic as : roads (like India and Netherlands does) , bridges, FURNITURE, bus stops, benches etc.

We could also take Thailand's idea to another level

In an entire country, there are water-dispensing machines. You put in any amount of money and you can refill your bottle with any amount of drinkable water (the fee is fixed only with bigger quantities). And we can make these machines in any shape and form also from the collected and recycled plastic.

This and having a reusable cup = environmental profit from stopping the production of plastic bottles. Profit from people buying water directly. And surely many others.



Such machines can be available in markets, gas stations, playgrounds, schools and basically anywhere. And if someone doesn't yet have a reusable cup, there can be an option to buy a cheap, recyclable cup from the machine. With a bin intended for collecting these cups also built into the machine.

Also remember, that if you live in a democratic country. **You can make a change**. Way towards it is clear, but uncertain - you can either make it in years or a month (with a good plan it mainly depends on luck).

But if you really crave a change here is what you do:

1. Develop a highly detailed plan. It must include a comprehensive description, implementation methods, potential obstacles and their solutions, projected outcomes, and a full breakdown of costs and profits. IMORTANT: you don't have to make it alone (explanation in the next point)

2. Find existing organizations directly related to the change you want to introduce. Generate enthusiasm among them and develop a plan together.

3. Clearly and compellingly present the key premises of the proposed change and rally support in the form of numerous and verifiable votes. The voices of both experts and ordinary residents matter—the latter are crucial to the voting outcome; if people desire this change, a politician seeking to implement it will see—well in advance—the potential for voter support.

4. Once you have a detailed plan and support in a form of votes, you can approach the relevant governing bodies directly. Ideally, start with the official responsible for the area you wish to change—for example, if you want schools to introduce classes on the importance of water, approach the Minister of Education.

Hopefully he will either see and support the vision or feel the pressure from the amount of supporters he could lose. Or it will lead to some other thought which eventually will lead to realization of the change.

See you in another on Earth Life part !

Through every part we will brighten the understanding of reality.

Continously, by diving deeper in it in first articles





Peace, Aeria Lilith



